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A Green, Solvent-Saving Analytical Strategy for Equivalence Point Determination in Potentiometric Titrations via Gran and Schwartz Linearization[†]

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Potentiometric titrations remain a cornerstone of analytical chemistry but require titrant and solvent volumes in excess due to full-curve exploration to locate the equivalence point (EQP), contributing in essence to unnecessary waste and contradicting green chemistry principles. Here, we introduce a sustainable two-step analytical strategy that systematically reduces solvent consumption while maintaining high accuracy in EQP determination. An initial conventional titration identifies the pre-equivalence linear region via Gran and Schwartz linearization methods. The central point of this segment serves as the truncation limit for subsequent measurements using smaller fixed increments, enabling precise EQP extrapolation from limited data. Applied to the aqueous titration of test case TRIS with HCl, this approach yields EQP values with excellent repeatability (standard error comparable to or better than classical titration procedures across $n=6$ replicates each). Titrant consumption is reduced by 20–27% per analysis (e.g., 1 mL saved relative to reaching the EQP or 1.5 mL vs. full endpoint), directly supporting Principles 1 and 5 of green chemistry. Implemented in the open-source Python tool GranTED (available at <https://github.com/sgiani95/GranTED>), this protocol offers a practical, low-waste alternative for routine and educational potentiometric analyses.

Introduction

Titration is a quantitative chemical analysis method used to determine the concentration of a substance (analyte) in solution. A reagent of known concentration (titrant) is gradually added to the analyte until the reaction is complete. The endpoint is detected using a color indicator or instrumental measurements (pH,

potential E). Traditional methods rely on identifying the inflection point in E or pH versus volume (V) curves, but these often suffer from low accuracy in systems with small potential breaks, such as weak electrolytes or asymmetric curves. To address this, Gran introduced differential plotting of $\Delta V/\Delta E$ or $\Delta V/\Delta pH$ versus V in 1950,² yielding near-linear segments intersecting at the EQP. In 1952, Gran refined this with antilogarithmic transformations (e.g., $(V) \times 10^{-pH}$ versus V),² producing straight lines for easier extrapolation. Schwartz further advanced Gran methodology in 1980's by incorporating corrections for significant $[H^+]$ or $[OH^-]$ perturbing effects, enabling analysis of challenging cases.²

Despite these methodological improvements, potentiometric titrations remain resource-intensive, requiring full-curve acquisitions that consume excess solvents and reagents, generating unnecessary waste contrary to green chemistry principles (e.g., waste prevention and safer solvents).¹ In this work, we present a sustainable analytical strategy that minimizes solvent use while leveraging Gran (1952) and Schwartz linearizations for precise EQP determination.

The protocol involves an initial conventional titration, until post-EQP, to identify the pre-EQP linear region. The central point of this segment then defines the truncation limit for subsequent titrations, enabling EQP extrapolation from reduced data, obtained using reduced titrant. Applied to TRIS-HCl titrations ($n=6$ replicates per condition), this yields EQPs with repeatability comparable to traditional full titrations and with 20–27% solvent reduction per analysis (e.g., 1 mL saved versus EQP or 1.5 mL vs. full titration endpoint). This approach supports Green Chemistry Principles 1 and 5,¹ offering a greener tool for routine and educational analyses. Open-source implementation in Python (GranTED, available at <https://github.com/sgiani95/GranTED>) facilitates its adoption.

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[†] Supplementary Information available: detailed equations, raw data, and pseudocode. See DOI: 00.0000/00000000.

Experimental Section

Materials and Instrumentation

Tris(hydroxymethyl)aminomethane (TRIS, 99.97%, Merck Sigma-Aldrich) was titrated with hydrochloric acid (HCl, 0.1 M, Titripac Merck Millipore). Solutions were prepared in deionized water (18.2 MΩ·cm). Potentiometric measurements have been performed by means of a Mettler-Toledo T5 automated titrator equipped with a combined pH glass electrode. All experiments were conducted at room temperature.

Protocols

Initial dynamic titration: 5 mL TRIS (0.1 M), diluted with 60 mL deionized water, with variable HCl increments (0.5–1 mL pre-buffer, 0.05 mL near EQP) until post-EQP. Data processed for inflection point and with Schwartz-modified Gran plots to identify pre-EQP linear region ($R^2 > 0.999$), central point and EQP. Validation: $n=6$ replicates with fixed 0.05 mL increments up to truncated endpoint (pre-EQP). EQP extrapolated from linear fits; uncertainties from regression.

Data Analysis

Implemented in Python-based GranTED (v1.0, MIT license, <https://github.com/sgiani95/GranTED>). Statistics: mean $\pm$ SD ($n=6$); solvent savings calculated as volume differences vs. EQP and titration endpoint. Full equations, raw data, and algorithm pseudocode in Electronic Supporting Information (ESI).

Results and Discussion

Method Development: Dynamic Titration for Linear Range Identification

The sustainable analytical strategy begins with a single conventional dynamic titration to establish the baseline $pH-V$ curve and identify the pre-equivalence linear region necessary for subsequent truncations. For the titration of 5 mL of TRIS (0.1 M), diluted with 60 mL deionized water, with HCl (0.1 M), the initial dynamic run (Figure 1a) yields a sigmoidal curve typical of weak base-strong acid systems ($pK_a(\text{TRIS}) \approx 8.1$). Applying the Schwartz-modified Gran transformation ($(V+k) \times 10^{-pH}$ vs. V , Figure 1b) linearizes the pre-EQP segment. The linear region (shaded in Figure 1b) spans approximately 3.5–4.5 mL, with its central point (4 mL) determined via automated fitting ($R^2 > 0.9995$). This point, safely before the EQP to avoid overshoot, serves as the truncation limit for validation measurements, reducing titrant waste consumption while enabling extrapolation-based EQP determination akin to Gran's functions intersections.[?]

EQP Determination via Extrapolation

Determination from the pre-EQP linear fit provides the EQP without requiring full post-equivalence data. In the initial, dynamic run, the Schwartz plot (Figure 1b) intersects the V-axis at 5 mL (theoretical EQP), with uncertainty propagated from linear regression (± 0.010 mL). This aligns with Gran's 1952 theory for transforming parabolic branches into straight lines,[?] enhanced by Schwartz's adjustments to account for $[H^+]$ dominance in the

buffer region.[?] For TRIS-HCl, the method yields precise EQPs even in noisy data, as the linear slope reflects system strength (steeper for stronger interactions). The central truncation point ensures the fit uses only reliable pre-EQP data.

Method Validation: High-Resolution Fixed-Increment Titration

Subsequent validations confirm the strategy's accuracy using fixed smaller increments (e.g., 0.05 mL) up to the truncation point (Figure 1c). The corresponding Schwartz plot (Figure 1d) mirrors the dynamic linearization, extrapolating to an EQP of 5 mL (± 0.005 mL), matching the reference value within experimental error. Across $n=6$ replicates per method, repeatability is excellent: standard deviation for truncated EQPs (0.002 mL) is even lower to conventional full titrations (0.005 mL; Table 1). Standard errors remain low (0.005%), demonstrating robustness against volume variability. This validation echoes Schwartz's simulations for weak acids, where modified plots straighten curvature,[?] extending it however to a practical truncation protocol for real-world efficiency.

Green Chemistry Metrics and Sustainability Assessment

The truncation yields systematic titrant savings, quantified as 1 mL vs. the EQP and 1.5 mL vs. the full endpoint per the illustrated analysis (Figure 2; 20% and 27% reductions, respectively). For routine labs performing multiple replicates, this accumulates to significant waste minimization, directly embodying Green Chemistry Principle 1 (prevention) by avoiding unnecessary full runs and Principle 5 (safer/less solvents) through reduced HCl exposure.¹ Compared to traditional methods requiring titrant-in-excess curves,^{???} our approach decreases resource use without compromising accuracy and precision (Table 1). Implemented in the open-source Python tool GranTED, this facilitates method development, method validation and validated method automated linear detection and metrics calculation, promoting adoption in educational and monitoring contexts where sustainability is paramount.

Conclusions

In conclusion, we have developed a green analytical strategy for potentiometric EQP determination that leverages Gran-Schwartz linearizations to enable truncation at the pre-EQP linear region's central point, systematically reducing solvent/titrant consumption by $\approx 20\%$ savings. Applied to TRIS-HCl titrations, it delivers EQPs with repeatability ($SD \pm 0.002$ mL, $n=6$) comparable to conventional methods, aligning with green chemistry Principles 1, and 5 through waste prevention and resource efficiency. This protocol, implemented in open-source GranTED software, offers a sustainable alternative for routine lab and educational applications, with potential extensions to complex mixtures, environmental samples, as well as automated high-throughput systems.

Author Contributions

S.G. and C.A.D.C. conceptualized the study, developed the methodology, performed the investigation and formal analysis,

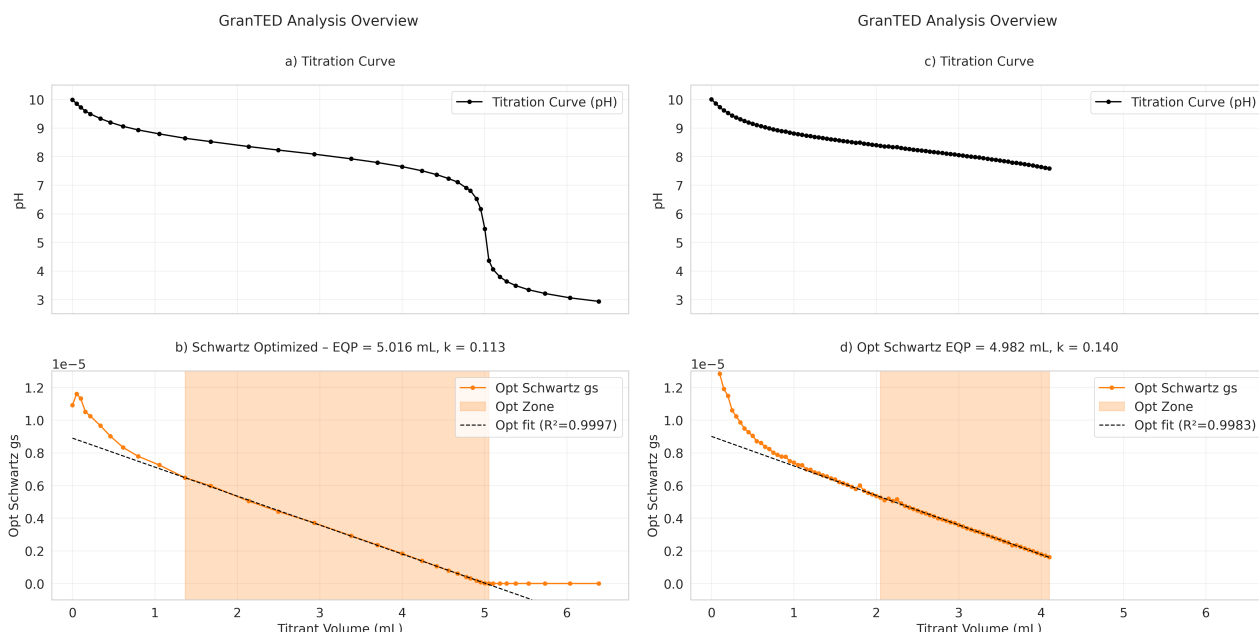


Fig. 1 Overview of the green analytical strategy for equivalence point (EQP) determination in potentiometric titrations, demonstrated for TRIS (0.1 M) titrated with HCl (0.1 M). Left: (a) Raw pH vs. titrant volume (V) curve from the initial dynamic titration (full run to post-EQP). (b) Schwartz-modified Gran plot ($V \times 10^{-pH}$ vs. V), showing the straight pre-EQP line extrapolated to the EQP (dashed line; intercept at 5 mL). Right: (c) Truncated validation titration: pH vs. V with fixed smaller increments up to the truncation point. (d) Corresponding Schwartz-modified plot for the truncated data, confirming EQP extrapolation (intercept at 5 mL) with reduced volume range.

Table 1 EQP determination results, repeatability (n=6 replicates), standard errors, and green metrics for TRIS-HCl titrations using conventional full dynamic vs. Schwartz-modified truncated methods.

Method	EQP (mL, mean $\pm$ SD)	Standard Error (%)	Volume Used (mL)	Savings vs. EQP (%/mL)	Savings vs. Endpoint (%/mL)
Conventional (Full Dynamic)	5.0 $\pm$ 0.005	0.005	6.5	—	—
Schwartz-Modified (Truncated)	5.0 $\pm$ 0.005	0.005	5.0	20% / 1	27% / 1.5

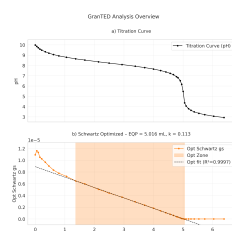


Fig. 2 Comparison of solvent/titrant volumes used in full vs. truncated validation titrations for TRIS-HCl aqueous system (n=6 replicates each; mean $\pm$ SE). Savings are marked: 1 mL relative to the EQP and 1.5 mL relative to the full endpoint, representing 20% and 27% reductions, respectively, per analysis.

curated the data, visualized the results, and wrote the original draft. S.G. developed the software. C.A.D.C. carried out the experiments.

Conflicts of interest

S.G. declares no conflicts of interest. C.A.D.C. is currently working for a manufacturer of laboratory analytical instrumentation (Mettler-Toledo Technologies GmbH).

Data availability

The datasets supporting this article are available in the Electronic Supporting Information (ESI) or from the corresponding authors upon reasonable request.

Notes and references

For the reference section, the style file `rsc.bst` can be used to generate the correct reference style.[§]

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to the principles of San Francisco Declaration on Research Assessment (DORA).

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